

THE UNIVERSITY OF HONG KONG
SCHOOL OF COMPUTING AND DATA SCIENCE
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data Mining
(Subclass B)

Date: December 9, 2024

Time: 6.30pm – 8.30pm

There are 4 questions in total. The 1st, 2nd and 3rd questions give 20/100 points each, while the 4th question gives 40/100 points each. Answer ALL questions. They are all **COMPULSORY**.

You should write your answers on the exam paper in the corresponding spaces left blank after the corresponding questions. Write only the final answers to the questions, without providing the intermediary calculations.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

University Number: _____

Brand and Type of Calculator: _____

Question 1 (PageRank, 20/100). Consider the directed graph G in Figure 1 which defines a Markov chain with four states.

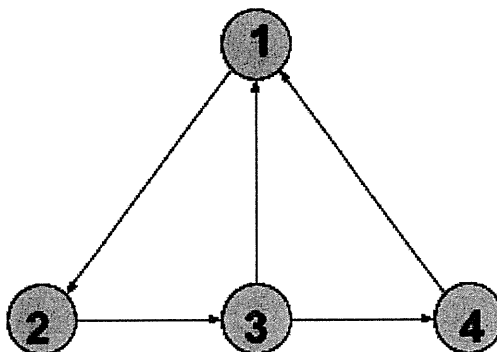


Figure 1. Input graph for PageRank.

- a) (8/100) Provide the PageRank vector of G : _____
- b) (6/100) Provide the vector computed at the end of the 1st iteration of PageRank with input G and initial vector $(1/4, 1/4, 1/4, 1/4)$: _____
- c) (6/100) Provide the vector computed at the end of the 2nd iteration of PageRank with input G and initial vector $(1/4, 1/4, 1/4, 1/4)$: _____

Question 2. (Apriori, PCY 20/100) We shall consider the problem of finding frequent itemsets and association rules in the set of baskets provided in Table 1, where each basket contains a set of items (e.g. set of products bought on an e-commerce website). For this question, use the following notation for the itemset counters: e.g. " $c_{24} = 2$ " specifies that the counter for itemset (2,4) has value 2. Write the counters in lexicographical order of their corresponding itemsets, e.g. $c_1=2$, $c_2=1$, $c_{13}=3$, $c_{24}=3$, $c_{31}=1$, $c_{123} = 2$, etc.

Baskets
1,2,3
2,4
3,4
1,3
2,5,7

Table 1.

- a) (4/100) What is the total number of counters (including all passes over the input data) of the Naive algorithm for finding all frequent itemsets (of any cardinality), with input the baskets in Table 1 and support threshold $s=2$? _____
- b) (4/100) Provide the counters and their values used by the A-priori algorithm at the end of the 2nd pass, with input the baskets in Table 1 and support threshold $s=2$:

- c) (4/100) Suppose the PCY algorithm uses the hash function $f(i,j) = (i + j) \% 3$, where $\%$ denotes the remainder of the integer division. Provide *only* the itemset counters and their values used by the PCY algorithm in the 2nd pass, with input the baskets in Table 1 and support threshold $s=2$ _____
- d) (4/100) What are the frequent itemsets in Table 1?

- e) (4/100) What is the support of the rule $2 \Rightarrow 3$ for the data in Table 1? _____
 What is the confidence of such a rule? _____

Question 3 (Clustering, 20/100). Recall that k clusters $C_1, \dots, C_k$ are well separated if and only if for every x, y, w, z with x, y in C_i , w in C_j and z in C_l , $j \neq l$, it holds that $d(x, y) < d(w, z)$. We are given a set of points $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$, let $k < n$. We denote with $d(x, y)$ the Euclidean distance between x and y . Consider the following algorithms:

A₁ (agglomerative clustering)

1. Start with n clusters $C_1, \dots, C_n$ with each cluster containing exactly one point x_i , i.e. $C_i = \{x_i\}$, $i=1, \dots, n$.
2. Until there are exactly k clusters:
 - a) Let C_i and C_j be two closest clusters (breaking ties arbitrarily), where the distance between two clusters C_i and C_j is defined to be the minimum distance between any two points x, y with x in C_i and y in C_j i.e. $d(C_i, C_j) = \min_{\{x \in C_i, y \in C_j\}} d(x, y)$.
 - b) Merge C_i and C_j that is, remove them from the list of the current clusters and add $C_i \cup C_j$ to the list of current clusters.

A₂ The k-means algorithm (with random initialization of the centroids).

A₃ The k-means algorithm (with centroid initialization computed by k-means++)

A₄ The k-means algorithm where the initial centroids are computed by k-means--.

Recall the k-means-- algorithm that we have seen in our course: It selects k centroids as follows: the first centroid is chosen uniformly at random from X . Then, let C be the current set of centroids with $|C| < k-1$. We add the point y in X to C , so that $d(y, C)$ is maximized. We iterate the previous step until $|C| = k$.

(20/100) Assuming that X can be partitioned into k well separated clusters. Which of the above algorithms can find a partition of X into k well separated clusters with probability 1? Specify them here in numerical order (e.g. A_2, A_3, A_4):

Question 4. (Decision Trees, 40/100). Consider the set X of examples shown in Table 2, where there are three binary features a, b, c , with each example being associated with the corresponding binary class value and an Id for each example.

Id	a	b	c	class
1	0	1	0	0
2	0	0	1	1
3	1	1	0	1
4	1	1	1	0

Table 2.

For this question, we use the greedy algorithm for building decision trees presented in our course, where the best split is determined using the Gini score. We use the following stopping conditions: the Gini value of the current node is 0, the maximum height of the tree is at most 2 (with the root being at height 0), or no further split is possible. The label of each leaf is the majority class in that leaf, with 0 being the default class. Moreover, given a decision tree T , we define the generalization error of T , denoted as $g(T)$ to be equal to:

$$g(T) = \alpha * \text{number of nodes in } T + \text{number of examples misclassified in } T, \quad \alpha = 2/7.$$

- a) (10/100) Build a decision tree on X , using the greedy algorithm presented in our course. Report in the space below, the decision tree built by the greedy algorithm with input X :
- b) (10/100) Provide a tree T such that $g(T)$ is min. among all possible decision trees on X .

c) (20/100) Determine a set $A \subseteq X$ of examples such that:

- i) Let Y_k be the set of examples where there are k copies of every example in A and one copy of every example in $X \setminus A$. For every $\gamma > 0$ there is k such that the generalization error of the tree T built by greedy with input Y_k is at least γ , i.e. greedy might produce arbitrarily "bad" decision trees.
- ii) $|A|$ is smallest among all set of examples satisfying i)

c₁) (5/100) Report the Ids of the examples in A in numerical order: _____

c₂) (5/100) Write down the generalization error $g(T)$ as a function of k , when T is built using the greedy algorithm with input Y_k _____

c₃) (5/100) Write down the generalization error $g(T)$ as a function of k , when T is a tree built on Y_k with minimum generalization error _____

c₄) (5/100) Write down the Gini of the splits on features a and b , with input Y_k as a function of k , respectively. We recall that the Gini of the split on a given feature is defined as $\frac{Gini(S)}{|S|} + \frac{Gini(X \setminus S)}{|X \setminus S|}$ where $S, X \setminus S$ is the partition of X obtained by splitting on that feature.

Gini when splitting on a _____

Gini when splitting on b _____

END OF PAPER

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For Rough Work Only

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For Rough Work Only